

Recursion + Backtracking + Trees — Complete DSA Sheet

All patterns covered | ~95 problems | Easy to Hard | LeetCode + GFG

1. RECURSION BASICS — Do This First

■ Master recursion before backtracking or trees. Every tree/graph problem uses recursion.

Foundation

- Factorial
- Fibonacci — Recursion then Memoization
- Power of X — Fast Exponentiation
- Reverse a String using Recursion
- Check if String is Palindrome — Recursion
- Sum of Digits
- Count Zeros in a Number
- Tower of Hanoi
- Print 1 to N without Loop
- Sum of N Natural Numbers

2. RECURSION ON ARRAYS (6 problems)

■ Apply recursion on arrays before moving to trees

Array Recursion

- Binary Search — Recursive
- Merge Sort — Recursive
- Quick Sort — Recursive
- Find Max in Array — Recursive
- Check if Array is Sorted — Recursive
- Reverse an Array — Recursive

3. BACKTRACKING FUNDAMENTALS (10 problems)

■ Choose → Explore → Unchoose. Always draw the recursion tree first.

Core Backtracking

- Subsets I — Power Set
- Subsets II — with Duplicates
- Permutations I
- Permutations II — with Duplicates
- Combination Sum I — Unlimited Use
- Combination Sum II — One Use Each
- Combination Sum III
- Letter Combinations of Phone Number
- Palindrome Partitioning
- Generate Parentheses

4. BACKTRACKING — HARD PROBLEMS (6 problems)

■ These are FAANG-level. Attempt only after basics are solid.

Hard Backtracking

- N-Queens (Hard) ■
- N-Queens II — Count Solutions (Hard) ■
- Sudoku Solver (Hard) ■
- Word Search I — Find Word in Grid
- Word Search II — Find All Words (Hard) ■
- Expression Add Operators (Hard) ■

5. BINARY TREE BASICS (10 problems)

■ Learn traversals first — everything else builds on top of them

Tree Traversals

- Inorder Traversal — Recursive + Iterative
- Preorder Traversal — Recursive + Iterative
- Postorder Traversal — Recursive + Iterative
- Level Order Traversal — BFS
- Zigzag Level Order Traversal

Basic Tree Problems

- Height / Max Depth of Binary Tree
- Diameter of Binary Tree
- Check if Tree is Balanced
- Mirror / Invert Binary Tree
- Same Tree / Symmetric Tree

6. BINARY TREE INTERMEDIATE (10 problems)

■ These are very common in interviews — must do all of them

Medium Tree Problems

- Path Sum I — Does Path Exist?
- Path Sum II — All Paths (Hard) ■
- Binary Tree Maximum Path Sum (Hard) ■
- Lowest Common Ancestor — LCA
- Binary Tree Right Side View
- Count Good Nodes in Binary Tree
- Flatten Binary Tree to Linked List
- Populating Next Right Pointers
- Serialize and Deserialize Binary Tree (Hard) ■
- Construct Tree from Inorder + Preorder

7. BINARY SEARCH TREE — BST (8 problems)

■ BST property: left < root < right. Use this to avoid full traversal.

BST Problems

- Search in BST
- Insert into BST
- Delete Node in BST
- Validate Binary Search Tree
- Lowest Common Ancestor in BST
- Kth Smallest Element in BST
- Inorder Successor in BST
- Two Sum in BST

8. TRIE — PREFIX TREE (5 problems)

■ Used for string search, autocomplete, prefix matching — very common in interviews

Trie Problems

- Implement Trie — Insert, Search, StartsWith
- Word Search II using Trie (Hard) ■
- Design Add and Search Words (Hard) ■
- Longest Word in Dictionary
- Replace Words

9. HEAP / PRIORITY QUEUE (8 problems)

■ Min-heap / Max-heap — Top K pattern. Know how to use PriorityQueue in Java.

Heap Problems

- Kth Largest Element in Array
- Kth Smallest Element in Array

- Top K Frequent Elements
- Sort Characters by Frequency
- K Closest Points to Origin
- Find Median from Data Stream (Hard) ■
- Merge K Sorted Lists (Hard) ■
- Reorganize String

10. STACK (8 problems)

■ Monotonic stack for next greater/smaller. Stack for expression evaluation.

Stack Problems

- Valid Parentheses
- Min Stack
- Evaluate Reverse Polish Notation
- Basic Calculator II
- Next Greater Element I
- Next Greater Element II — Circular
- Daily Temperatures
- Largest Rectangle in Histogram (Hard) ■

11. LINKED LIST (10 problems)

■ Very common in interviews — slow/fast pointer is key pattern

Linked List

- Reverse Linked List
- Merge Two Sorted Lists
- Linked List Cycle — Detect
- Linked List Cycle II — Find Start
- Middle of Linked List
- Remove Nth Node from End
- Add Two Numbers
- Copy List with Random Pointer (Hard) ■
- Reverse Nodes in K Group (Hard) ■
- LRU Cache (Hard) ■

SUMMARY

#	Topic	Problems	Hard
1	Recursion Basics	10	0
2	Recursion on Arrays	6	0
3	Backtracking Fundamentals	10	0
4	Backtracking Hard	6	5
5	Binary Tree Basics	10	0
6	Binary Tree Intermediate	10	3
7	BST	8	0
8	Trie	5	2
9	Heap / Priority Queue	8	2
10	Stack	8	1
11	Linked List	10	3
TOTAL		91	16

Recommended Order: Recursion Basics → Recursion on Arrays → Backtracking → Binary Tree → BST → Heap → Stack → Linked List → Trie

■ Red = Hard | ■ = Not done | Trees appear in almost every interview — do not skip!